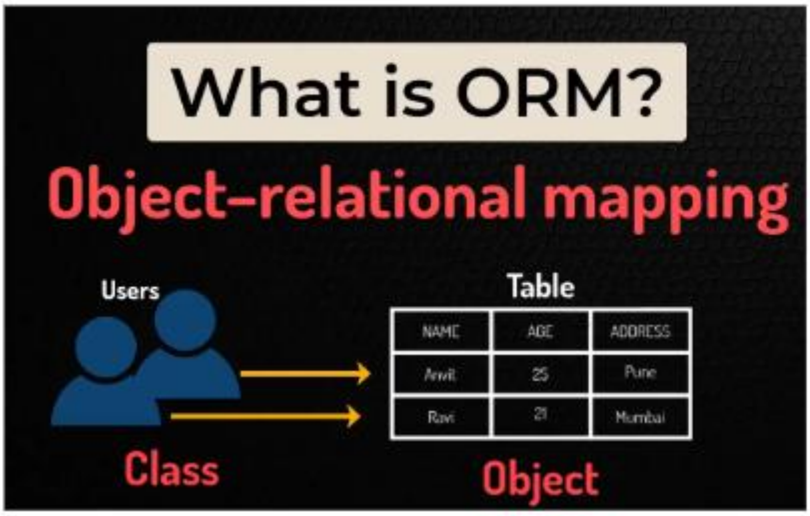
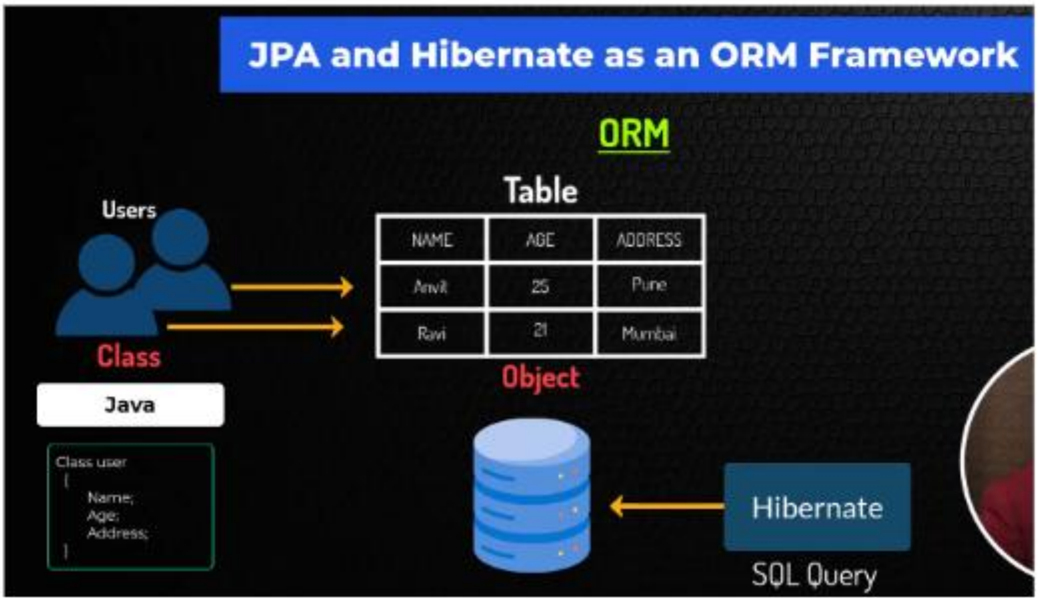
JPA and Hibernate as ORM Framework





What is ORM (Object-Relational Mapping)?

- ***Definition****:* ORM is a technique that allows developers to map Java objects to database tables, making it easier to work with databases in an object-oriented manner.
- ***How It Works****:*
  - Java classes (representing entities) are mapped to database tables.
  - Each instance of a class corresponds to a row in the table.
  - This mapping allows developers to manipulate database records as if they were working with simple Java objects.

JPA (Java Persistence API):

- ***Definition:*** JPA is a specification provided by Java for managing relational data in Java applications.
- ***Key Points:***
  - ***Object-Relational Mapping:*** It defines how to map Java objects to database tables.
  - ***Implementations:*** Hibernate and TopLink are popular implementations of the JPA specification.

Hibernate:

- ***Definition:*** Hibernate is a popular ORM framework that implements the JPA specification, providing a powerful and flexible way to work with relational databases in Java.
- ***How It Works:***
  - Hibernate automatically generates SQL queries based on the mapping defined between Java classes and database tables.
  - These queries are then executed using JDBC (Java Database Connectivity) to interact with the database.

Key Benefits:

- ***Automated SQL Generation:*** Hibernate generates SQL queries automatically, reducing the need for manual query writing.
- ***Ease of Data Manipulation:*** Developers can work with database data as Java objects, simplifying code and improving maintainability.

Summary:

- ***ORM (Object-Relational Mapping):*** Maps Java classes to database tables.
- ***JPA (Java Persistence API):*** A specification that defines how ORM should be implemented in Java.

Hibernate: A widely-used implementation of JPA that automates database interactions, making Java development more efficient and less error-prone.